

M. Thufail Alwannabil Samas

Surabaya, East Java, Indonesia | +62 877-6313-3209 | mthufailalwannabils@gmail.com

LinkedIn: linkedin.com/in/mthufailsamas | GitHub: github.com/mthufailsamas

Portfolio: mthufailsamas.github.io

SUMMARY

Mathematics graduate and BNSP-certified Data Scientist with experience in data analysis, machine learning, time-series analysis, and Python/PostgreSQL data pipelines. Developed monitoring, forecasting, and classification workflows, with a first-author peer-reviewed publication on XGBoost-based rainfall forecasting.

PROFESSIONAL EXPERIENCE

PT Aplikasi Lintasarta

Surabaya, Indonesia

Junior Programmer

January 2026 – April 2026

- Developed a project-stage DWDM optical monitoring and assurance analytics system for approximately 400 directional sensors, integrating 5-minute PRTG telemetry with ENIMS network metadata to identify operational failures, persistent signal changes, gradual degradation, and sensor priorities.
- Built an incremental Python and PostgreSQL time-series pipeline for ENIMS metadata synchronization, PRTG telemetry collection, gap repair, processed-observation generation, event detection, paired-sensor correlation, priority ranking, and API-served dashboard outputs.
- Implemented data-quality and failure-detection logic for missing or empty observations, low optical-power values, downtime, and zero coverage, collapsing adjacent 5-minute failure buckets into contiguous events while excluding invalid observations from statistical analysis.
- Developed sensor-adaptive change-point and gradual-degradation detectors using robust 6-hour medians, penalized PELT segmentation, Theil-Sen slopes, and serial-correlation-corrected Mann-Kendall tests, with regime and step-dominance checks to distinguish persistent signal changes from telemetry noise.
- Built a Python-backed HTML/CSS/JavaScript dashboard with Operations Board, Event Explorer, and Management Overview views for active failures, statistical events, paired-sensor evidence, investigation charts, and current sensor-priority ranking.

SMP GIKI 3 Surabaya

Surabaya, Indonesia

Coding and AI Teacher

August 2025 – December 2025

- Taught Python fundamentals to Grade 8 and 9 students over 5 months, covering variables, data types, operators, conditionals, loops, and programming logic through practical coding exercises.
- Introduced foundational AI concepts using Teachable Machine, guiding students through datasets, model training, and image classification through hands-on experimentation.
- Designed lesson materials, coding exercises, quizzes, and sample code from scratch, using practical programming activities and assessments to evaluate student understanding.

Stasiun Meteorologi Maritim Kelas II Tanjung Perak Surabaya (BMKG)

Surabaya, Indonesia

Data Science Intern

February 2024 – June 2024

- Developed a one-page extreme-weather early warning dashboard using HTML, CSS, JavaScript, PHP, MySQL, and API-based periodic updates for BMKG's internal forecasting workflow.
- Designed monitoring views for current weather conditions, warning indicators, tables, charts, and filters, consolidating key information into a single operational interface.
- Processed and compared BMKG cloud-prediction imagery with actual Himawari IR satellite imagery using Python, supporting descriptive validation of cloud movement and temperature patterns.
- Developed a 3-layer BiLSTM rainfall forecasting model using 7-day sequences and an 80:20 chronological split, achieving MAAPE of 0.8073 and RMSE of 10.2734 mm.

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA)

Surabaya, Indonesia

Programming Teaching Assistant

August 2022 – July 2024

- Guided undergraduate students across 4 semesters of Basic Programming and Programming courses in MATLAB, Python, and computational problem-solving.
- Provided hands-on guidance in debugging, data handling, and algorithmic thinking, helping students translate mathematical and computational concepts into working code.

- Developed introductory machine-learning materials covering 3 algorithms – Linear Regression, K-Nearest Neighbors, and K-Means clustering – and provided structured technical feedback on student assignments.

EDUCATION

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA)
Bachelor of Mathematics (S.Mat.)

Surabaya, Indonesia
August 2021 – January 2025

- **GPA:** 3.46/4.00
- **Undergraduate Thesis:** Rainfall Forecasting in Sumenep Regency Using XGBoost and Grid Search.
- **Academic Highlight:** Completed the 4-year bachelor's program in 7 semesters (3.5 years).
- **Relevant Coursework:** Artificial Intelligence, Programming, Mathematical Statistics, Statistical Methods, Multivariate Analysis, and Applied Linear Algebra.

TECHNICAL SKILLS

- **Programming & Query:** Python, SQL, R, MATLAB, PHP
- **Data Engineering:** PostgreSQL, MySQL, Data Ingestion, Data Validation, Data Transformation, Data Pipelines
- **Libraries & Frameworks:** NumPy, Pandas, Scikit-learn, XGBoost, Statsmodels, TensorFlow/Keras, Matplotlib, Seaborn
- **Analytics & Modeling:** Data Analysis, Data Preprocessing, Feature Engineering, Statistical Modeling, Machine Learning, Deep Learning, Model Evaluation, Time Series Analysis & Forecasting
- **Development & Visualization:** Jupyter Notebook, Git, GitHub, REST API, HTML, CSS, JavaScript, Dashboard Development

PROJECTS

Rainfall Forecasting with XGBoost and Grid Search
Built an XGBoost next-day rainfall forecasting workflow using 1-day lagged BMKG weather features, fold-wise Min-Max scaling, 10-fold TimeSeriesSplit, and Grid Search, achieving MAAPE of 0.9152 and RMSE of 11.9566 mm.

Thyroid Cancer Recurrence Classification with XGBoost, Information Gain, and Grid Search
Built an XGBoost recurrence-classification workflow using the 383-record UCI dataset, fold-wise Information Gain, 10-fold cross-validation, and Grid Search, achieving 97.25% accuracy, 95.37% F1-score, 98.05% specificity, and 98.60% ROC-AUC.

Rainfall Forecasting with BiLSTM and Grid Search
Built a 3-layer BiLSTM next-day rainfall forecasting workflow using 7-day multivariate sequences, an 80:20 chronological split, Min-Max normalization, and Grid Search, achieving MAAPE of 0.8073 and RMSE of 10.2734 mm.

Rice Price Forecasting with ARIMA and Walk-Forward Validation
Built two ARIMA forecasting models for 60 months of East Java medium and premium rice prices using ADF-based differencing, residual diagnostics, and expanding-window walk-forward validation, achieving MAPE of 2.081% and 1.8174%, respectively.

PUBLICATIONS

Prediksi Curah Hujan di Kabupaten Sumenep Menggunakan Metode Extreme Gradient Boosting (XGBoost) dan Algoritma Grid Search
First Author | Peer-Reviewed Article — *MIND Journal*, Vol. 11, No. 1, pp. 30 – 43, June 2026
DOI: 10.26760/mindjournal.v11i1.30-43

CERTIFICATIONS

- **Data Scientist** — Badan Nasional Sertifikasi Profesi (BNSP) | September 2024 – September 2027